The Egyptian Family Health Companion (3elty)

BitCare Team

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1. User Stories

Priority Scale

- **Highest —** Core system function; the app cannot fulfill its purpose without it.
- **High —** Critical safety or key differentiating feature.
- **Medium —** Valuable enhancement; improves experience but not a blocker.

Story Points Baseline

- Filling a structured form — such as the medical profile with chronic conditions, allergies, and blood type — is our baseline reference, rated 3 story points.
- Adding an appointment to the shared calendar is a second reference point — a standard form with scheduling logic, also rated 3 story points.
- All other stories are estimated relative to these two baselines — simpler tasks fall below 3, more complex ones scale up accordingly.

Acceptance Criteria:

- **Definition:** The specific conditions that a software product must meet to be accepted by a user, customer, or other stakeholder.
- **Purpose:** They define the "boundaries" of a User Story and provide a clear checklist to determine when a feature is **Done**.
- **Format:** We use the **Given-When-Then** (Gherkin) syntax to ensure each criterion is testable, objective, and written in a single, clear line of logic.

1.1 Authentication & Family Account Epic

As a family admin, I want to create a family account with my name, phone number, and password, so that I can manage all family members' health in one place.

The family admin sets up the central account that acts as the main hub for all health management. This is the very first step a new user takes — without it, no other feature in the app is accessible.

Highest

Every other feature — profiles, calendars, medication tracking — is inaccessible until a family account exists. It is the foundational entry point of the entire application.

3 Story Points

The form covers 3 fields (name, phone, password) with basic validation and secure password hashing.

<u>Acceptance Criteria:</u>

- Given a new user opens the app for the first time, When they tap "Create Family Account", Then they see a form with Name, Phone Number, and Password fields
- Given the user fills all fields correctly, When they tap "Create", Then the account is created and they are redirected to the home dashboard
- Given the user enters an already-registered phone number, When they tap "Create", Then they see the error: "This phone number is already registered"
- Given the user leaves any required field empty, When they tap "Create", Then each empty field shows a red error message
- Given the password is less than 8 characters, When they tap "Create", Then they see: "Password must be at least 8 characters"

As a family admin, I want to add family members with their name, age, and profile type (child / elderly / pregnant / chronic / adult), so that each member gets the right health modules.

After creating the account, the admin populates it by adding each family member and assigning them a profile type. Each type unlocks a tailored set of health modules for that member.

Highest

The app is role-based — without members being registered and typed, no health module can be displayed for anyone. This story directly enables all downstream features for every family member.

3 Story Points

Involves a form with a role dropdown and input validation, plus logic to map each profile type to the correct visible modules.

<u>Acceptance Criteria:</u>

- Given the admin is logged in, When they tap "Add Family Member", Then they see a form with Name, Age, and Profile Type fields
- Given the admin selects profile type "Child", When the member is saved, Then the Child Module is activated for this member
- Given the admin selects profile type "Elderly", When the member is saved, Then the Elderly Module is activated
- Given the admin selects profile type "Pregnant", When the member is saved, Then the Pregnancy Module is activated for this member
- Given the admin selects profile type "Chronic", When the member is saved, Then the Chronic Disease Module is activated
- Given a member is added successfully, When the admin views the family dashboard, Then the new member appears with their profile type icon

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As a family member, I want to log in with my phone number and PIN, so that I can access my profile securely without a complex password.

Any family member can log in using their phone number and a short PIN, providing simple and secure access — especially important for elderly users or less tech-savvy members of the family.

Highest

Authentication is the gateway to all personal health data. Without it, no personalized feature — medications, vitals, calendar — is accessible. It is a non-negotiable system prerequisite.

2 Story Points

PIN-based login is a two-field form with backend verification and session token handling.

<u>Acceptance Criteria:</u>

- Given a registered user opens the app, When they enter their phone number and correct PIN, Then they are logged in and see their personal profile
- Given a user enters a wrong PIN 3 times, When the third attempt fails, Then the account is locked for 5 minutes and they see a warning
- Given an elderly user opens the app, When they see the login screen, Then the PIN input uses large buttons.
- Given a logged-in user closes the app, When they reopen it within 24 hours, Then they remain logged in (session persistence)

1.2 Child Module Epic

As a parent, I want to see my child's full vaccination schedule based on Egypt's Ministry of Health calendar, so that I know exactly which vaccines are coming and when.

Parents can view a pre-built vaccination schedule aligned with Egypt's Ministry of Health guidelines, showing which vaccines are due and when — replacing the need to consult external sources or paper booklets.

Highest

This is the foundational view for child health management. All vaccine-related activity anchors on it — parents must see the schedule before they can record completions or receive reminders.

3 Story Points

Requires loading the MOH vaccine schedule, mapping it to the child's birthdate, and rendering a timeline view.

Acceptance Criteria:

- **Given** a parent has registered their child's birth date, **When** they open the vaccination module, **Then** they see a full timeline based on Egypt's Ministry of Health calendar.
- **Given** the child is 2 months old, **When** the parent views the schedule, **Then** the system highlights the "2-month" vaccines (e.g., Polio, DTaP) as the current priority.
- **Given** the user is looking at the schedule, **When** they tap on a specific vaccine name, **Then** the system displays a brief description of what the vaccine protects against.
- **Given** the parent scrolls through the timeline, **When** they view future dates, **Then** the system displays the exact expected dates for upcoming vaccines based on the birth date.

As a parent, I want to mark a vaccine as received and record the clinic name and date, so that I have a permanent digital record replacing the paper booklet.

Parents can check off a vaccine as administered, recording the clinic and the exact date. This creates a permanent, verifiable digital record that replaces the traditional paper vaccination booklet.

Highest

This story closes the loop on the vaccination schedule — seeing what is due is only useful if completed vaccines can be recorded. It is a core function for the child health module.

5 Story Points

Requires updating schedule state, storing clinic and date metadata per vaccine, and reflecting changes in the shared calendar.

Acceptance Criteria:

- **Given** a vaccine is listed as "Due", **When** the parent taps "Mark as Received" and enters the clinic name and date, **Then** the status updates to "Completed".
- **Given** a vaccine has been marked as completed, **When** the parent views the child's profile, **Then** the system shows a permanent digital record with the clinic and date metadata.
- **Given** the user is filling the completion form, **When** they leave the "Date" field empty, **Then** the system shows a red error message and prevents saving.
- **Given** a vaccine is updated to "Completed", **When** the user checks the shared family calendar, **Then** the completion event is automatically reflected there for all members.

As a parent, I want to log my child's weight and height regularly, so that I can track their growth against WHO percentile charts.

Parents can record periodic weight and height measurements for their child. The app plots these against WHO growth percentile charts so parents can monitor healthy development over time.

Medium

Growth tracking is a valuable parenting tool but not a critical safety function. The app delivers full value for emergency care, medications, and appointments without it.

5 Story Points

Requires data entry forms, a time-series chart rendering WHO percentile bands, and percentile calculation.

Acceptance Criteria:

- **Given** a parent wants to track growth, **When** they enter the child's current weight and height and tap "Save", **Then** the data is stored with the current timestamp.
- **Given** new growth data has been entered, **When** the parent opens the growth chart, **Then** the system plots the new point against the standard WHO percentile bands.
- **Given** the user enters a measurement, **When** the system processes the data, **Then** it automatically calculates and displays the child's current percentile (e.g., 50th percentile).
- **Given** the user views the growth history, **When** they look at the time-series chart, **Then** they can see the growth trend line connecting all previous measurements.

1.3 Pregnant Woman Module Epic

As a pregnant woman, I want to see the required prenatal tests for each trimester with reminders, so that I never miss a critical blood test, ultrasound, or glucose screening.

Pregnant users can view a structured checklist of required blood tests, ultrasounds, and screenings organized by trimester, with automated reminders before each deadline.

Highest

Missed prenatal screenings can have serious consequences for mother and baby. This is a core safety feature for the pregnancy module and is central to the app's value for this user type.

3 Story Points

Requires a trimester-based test schedule mapped to the user's due date and a reminder scheduler.

Acceptance Criteria:

- **Given** a pregnant user has set her due date, **When** she opens the prenatal tests screen, **Then** she sees a checklist categorized by the 1st, 2nd, and 3rd trimesters.
- **Given** a specific test is listed as pending, **When** the user taps "Mark as Completed", **Then** the test status updates to "Done" and is saved in her history.
- **Given** an upcoming test deadline is approaching, **When** the system reaches the reminder time, **Then** an automated push notification is sent to the user.
- **Given** the user is in the 2nd trimester, **When** she views the checklist, **Then** the system highlights the 2nd trimester screenings as the current priority

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As a pregnant woman, I want to log my daily medications (folic acid, iron, calcium) and mark them as taken, so that I maintain a consistent prenatal medication routine.

Pregnant users can track their prescribed prenatal supplements on a daily basis, marking each one as taken to maintain a consistent routine for medications that are clinically critical throughout pregnancy.

Highest

Skipping prenatal supplements like folic acid or iron can cause birth defects or anaemia. This is a core daily safety feature for the pregnancy module.

3 Story Points

Involves a recurring daily medication list with a one-tap confirmation mechanism.

Acceptance Criteria:

- **Given** a user needs to track daily supplements, **When** she taps the icon for Folic Acid or Iron, **Then** the specific medication is marked as "Taken" for that day.
- **Given** all daily medications have been confirmed, **When** the final supplement is marked as taken, **Then** the system displays a visual "Success" confirmation state.
- **Given** the user has completed her previous day's routine, **When** the time reaches 12:00 AM, **Then** the medication checklist resets to "Pending" for the new day.
- **Given** the user marks a medication as taken, **When** she checks her history, **Then** the system shows a timestamped record of her adherence for that day.

As a pregnant woman, I want to see a color-coded list (Green for safe, Red for unsafe) of common local dishes, so that I can make quick decisions while cooking or ordering food.

A food safety reference showing common Egyptian local dishes marked as safe or unsafe for pregnant women, helping users make fast dietary decisions without needing to research each dish individually.

Medium

A useful convenience feature but not a core health function. The app delivers full prenatal care value — medications, tests, calendar — without it. Rated Medium as a quality-of-life enhancement.

2 Story Points

A static curated list with a safe/unsafe display and a basic search filter.

Involves a recurring daily medication list with a one-tap confirmation mechanism.

Acceptance Criteria:

- **Given** the user is browsing local dishes, **When** she views a dish like "Faseekh", **Then** the system displays a **Red** indicator to signify it is unsafe during pregnancy.
- **Given** the user is searching for a safe meal, **When** she types a dish name in the search bar, **Then** the list filters to show only matching results with their safety status.
- **Given** the user has selected a dish from the guide, **When** she taps on its name, **Then** the system shows a brief safety note explaining why it is safe or unsafe.

- **Given** the user wants to filter the guide, **When** she selects a category like "Seafood", **Then** the system displays all related Egyptian dishes with their respective Green/Red color codes

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1.4 Chronic Patient Module Epic

As a chronic patient, I want to log my daily vital signs (blood sugar, blood pressure), so that I can track my health trends over time.

Chronic patients can log key daily vitals such as blood sugar and blood pressure. The app stores these over time and displays trend charts, helping patients and doctors identify patterns in their condition.

Highest

Daily vital logging is the core data-entry function for chronic disease management. All downstream features — trend analysis, auto-reports, and threshold alerts — depend on this data existing.

3 Story Points

Involves forms for multiple vital types, persistent time-series storage, and a basic trend chart.

<u>Acceptance Criteria:</u>

- **Given** the patient is on the Vitals screen, **When** they enter a value for blood sugar or blood pressure, **Then** the system shall validate the input and save it with a timestamp.
- **Given** historical data exists, **When** the user accesses the trends section, **Then** the system shall render a time-series chart displaying the health patterns.
- **Given** an empty or invalid input, **When** the user attempts to save, **Then** the system shall display a corrective error message and prevent data submission.

As a chronic patient, I want a medication adherence tracker so that I don't miss any doses of my life-saving medicine.

Chronic patients can configure their medication schedule and mark each dose as taken. The tracker monitors adherence over time, flags missed doses, and provides a history that can be shared with physicians.

Highest

For chronic patients, missing medication can be life-threatening. This is the daily touchpoint that underpins the remote missed-dose alert sent to family admins — making it a primary system function.

3 Story Points

Requires a configurable medication schedule, recurring reminders, a one-tap confirmation, and missed-dose detection logic.

<u>Acceptance Criteria:</u>

- **Given** the medication configuration screen, **When** the user sets a schedule, **Then** the system shall trigger recurring reminders at the specified times.

- **Given** a pending medication dose, **When** the user performs a "one-tap" confirmation, **Then** the system shall update the status to "Taken" in the history.
- **Given** a dose remains unconfirmed after its scheduled time, **When** the tracking job runs, **Then** the system shall flag it as "Missed" and notify the family admin.

As a chronic patient, I want an auto-generated monthly clinical summary so that I can show it to my doctor during visits.

At the end of each month, the app automatically compiles the patient's logged vitals, medication adherence rate, and notable trends into a structured clinical summary ready to present to a doctor.

Highest

This is a flagship feature for the chronic patient module — it bridges daily self-monitoring and professional care, directly improving the quality of doctor-patient consultations.

5 Story Points

Requires aggregating a month of vitals and adherence data, computing statistics, generating a formatted document, and enabling sharing.

Acceptance Criteria:

- **Given** a month of logged data, **When** the user requests a summary, **Then** the system shall aggregate all vitals and adherence rates into a structured report.
- **Given** the summary is compiled, **When** the user selects the export option, **Then** the system shall generate a formatted PDF document.
- **Given** the PDF is ready, **When** the user taps "Share", **Then** the system shall open the native share sheet for external distribution.

As a chronic patient, I want to receive an alert when my vitals exceed a dangerous threshold, so that I can take immediate action even if I am not fully aware of the severity

When a vital sign logged by the patient crosses a predefined danger threshold — such as very high blood pressure or blood sugar — the patient receives an immediate push notification to take appropriate action.

Highest

Vital-threshold alerts are a core safety mechanism for chronically ill individuals. This feature ensures that critical health conditions are not overlooked and enables timely self-intervention to reduce the risk of complications.

5 Story Points

Requires threshold configuration per member, a real-time monitoring hook on vital submissions, and a notification pipeline.

Acceptance Criteria:

- **Given** the threshold settings, **When** a user defines a "danger zone" for their vitals, **Then** the system shall store these limits for real-time monitoring.
- **Given** a new vital entry, **When** the value exceeds the predefined limit, **Then** the system shall immediately dispatch a push notification alert.

- **Given** the alert is triggered, **When** the user opens it, **Then** the system shall provide immediate actionable medical instructions.

1.5 Elderly User Module Epic

As an elderly user, I want to confirm that I took my medication with one tap, so that my family knows I didn't miss a dose.

Elderly users see their scheduled medications on screen and confirm each one with a single tap. The interface is deliberately minimal — designed for users who may struggle with technology.

Highest

This is the elderly user's primary daily interaction with the app. Without it, the family admin's missed-dose alert system has no trigger. It is a core function that underpins the entire medication safety chain.

2 Story Points

A single large confirmation button — deliberately minimal UI and logic.

<u>Acceptance Criteria:</u>

- Given a scheduled medication, when the user taps the confirm button, then the system records the dose as taken with a timestamp.
- Given the user taps the confirm button, when the action succeeds, then a clear confirmation message is displayed.
- Given a medication is confirmed, when the record is saved, then all linked family members receive a notification.
- Given a medication has already been confirmed, when the user attempts to confirm again, then the system prevents or flags the duplicate action.

As an elderly user, I want to press a panic button that sends my GPS location to all family members, so that my family can respond immediately in an emergency.

A large, always-visible panic button on the elderly user's home screen. When pressed, it immediately broadcasts the user's current GPS coordinates to all registered family members via push notification.

High

A critical safety feature, but rated High rather than Highest because it is an emergency-use fallback rather than a daily core function. The daily medication confirmation is more operationally central.

8 Story Points

Requires real-time GPS acquisition, multi-recipient push notification dispatch, a prominent UI element, and fallback behavior when location is unavailable.

<u>Acceptance Criteria:</u>

- Given the user presses the panic button, when it is activated, then an emergency alert is triggered immediately.
- Given an emergency alert is triggered, when it is processed, then the user’s current GPS location is included.
- Given an alert is sent, when family members are registered, then all of them receive the alert via notification channels.

- Given the panic button is pressed, when the alert is successfully sent, then the user sees a confirmation message.
- Given the panic button is pressed accidentally, when it occurs within a short cancellation window, then the user can cancel the alert.

As a family admin living abroad, I want to receive a notification if my elderly parent missed their medication, so that I can call and check on them remotely.

If an elderly member has not confirmed taking their scheduled medication within the expected time window, the admin receives a push notification — critical for families where the admin lives away from their elderly relatives.

Highest

Medication non-adherence in elderly patients is a leading cause of preventable hospitalization. This remote alert directly delivers on the app's promise to connect distant caregivers to vulnerable relatives.

5 Story Points

Requires tracking confirmation timestamps, a scheduled job to detect missed doses, and a notification delivery system.

Acceptance Criteria:

- Given a medication schedule exists, when the user does not confirm within the allowed time window, then the system marks the dose as missed.
- Given a dose is marked as missed, when the status updates, then the family admin receives a notification.
- Given a missed dose notification is received, when the admin views it, then it includes the medication name, missed time, and user identity.
- Given the admin is in a different time zone, when the notification is displayed, then the time is shown in the admin's local time zone

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1.6 Medical Records Epic

As a family member, I want to fill in my medical profile including chronic conditions, allergies, and blood type, so that any doctor can quickly understand my medical history.

Each member completes a structured medical profile capturing chronic conditions, known allergies, and blood type. This profile is the source of truth that doctors can reference during any visit.

Highest

The medical profile is foundational health data. It powers the PDF summary, informs the doctor during visits, and enables the app to tailor alerts. Without it, the platform has no clinical value.

3 Story Points

A multi-field form with structured data entry — multi-select for conditions and allergies, dropdown for blood type — and persistent storage.

Acceptance Criteria:

- Given a family member opens their profile, When they tap "Medical Profile", Then they see fields for: Blood Type, Chronic Conditions, Allergies, Past Surgeries

- Given all fields are filled and saved, When a doctor views the QR emergency card, Then they see the complete profile summary

- Given a member has no known allergies or chronic diseases , When they leave the allergies and chronic conditions fields empty, Then it shows "No known / NA " on the summary

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As a family member, I want to upload a photo of my lab result or prescription, so that all my medical documents are stored digitally in one place.

Family members can photograph paper lab results or prescriptions and upload them to their digital health record. All documents are stored centrally and are instantly retrievable for any future appointment.

Highest

Egypt's healthcare system is heavily paper-based. Centralizing records digitally is one of the app's core differentiators and directly addresses a critical daily pain point for users.

8 Story Points

Requires camera/gallery integration, image compression, secure cloud storage, document indexing, and a retrieval UI.

<u>Acceptance Criteria:</u>

- Given a member taps "Upload Document", When they choose from camera or gallery, Then the image is uploaded and saved to their profile

- Given a document is uploaded, When the member views their documents list, Then they see a thumbnail with upload date and a label they added
- Given the upload fails due to network, When the error occurs, Then they see: "Upload failed — please check your connection and try again

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As a family member, I want to generate a one-page PDF summary of my medical record, so that I can share it with any new doctor at a clinic visit.

A family member can export their complete medical profile as a clean, portable one-page PDF that can be handed to any new doctor during a clinic visit — eliminating the need to repeat history verbally.

High

An important portability feature, but it depends on the medical profile already being filled in. It is rated High because it enhances usability rather than enabling core data entry.

5 Story Points

Requires PDF generation from structured profile data, formatting into a medical layout template, and download/share functionality.

<u>Acceptance Criteria:</u>

- Given a member taps "Generate Summary", When the PDF is created, Then it includes: Name, Age, Blood Type, Conditions, Allergies, Current Medications.

- Given the PDF is generated, When the member taps "Share", Then the phone's native share sheet opens with the PDF attached.

- Given the medical profile is incomplete, When they try to generate the PDF, Then they see: "Please complete your medical profile before generating a summary".

1.7 Family Calendar Epic

As a family admin, I want to see all family appointments, upcoming vaccines, and tests in one shared calendar, so that I can plan and coordinate everyone's health events at a glance.

A unified calendar view that aggregates every appointment, vaccine due date, and lab test deadline for all family members in one place, allowing the admin to coordinate the family's health schedule.

Highest

The shared calendar is the coordination hub that gives the app its 'family health manager' identity. Removing it would eliminate a core value proposition of the platform.

8 Story Points

Requires aggregating data across multiple profiles and event types, building a calendar UI with filtering, and handling real-time updates.

<u>Acceptance Criteria:</u>

- Given the admin opens the calendar, When it loads, Then they see events from ALL family members color-coded by member.

- Given a vaccine is due next week, When the admin views the calendar, Then the vaccine appointment appears automatically (no manual entry needed).

- Given the admin taps any event, When tapped, Then they see: Member name, Event type, Date, Time, Notes.

As a family member, I want to add a doctor appointment to the shared family calendar, so that the whole family is aware and I receive a reminder before it.

A family member can schedule a new doctor appointment, have it appear on the shared calendar, and receive an automated reminder — keeping the whole family informed and reducing missed visits.

Highest

This input story is inseparable from the calendar's usefulness — without it, the shared calendar remains empty. It is a core feeding action for one of the app's primary features.

3 Story Points

A date/time form that writes to the shared calendar and schedules a push reminder.

Acceptance Criteria:

- Given a family member taps "Add Appointment", When they fill: Doctor Name, Specialty, Date, Time, Notes, Then it is saved and appears on the shared calendar.
- Given an appointment is 24 hours away, When the system checks, Then both the member and admin receive a reminder notification.
- Given an appointment is added, When the admin views the calendar, Then it appears with the correct member's color coding

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2.Use Case List

2.1 Use Case: Add Family Member

- **Actor:** Family Admin.
- **Description:** Admin adds a new member and assigns their health profile.
- **Main Flow:**
    1. Admin enters member's name, age, and profile type (e.g., Child, Elderly).
    2. System validates data and unlocks specific health modules for that role.
    3. System confirms the new profile is created.

2.2 Use Case: Track Child Vaccination

- **Actor:** Parent / Family Admin.
- **Description:** Parent views the official schedule and marks vaccines as completed.
- **Main Flow:**
    1. Parent views the vaccination timeline based on MOH guidelines.
    2. Parent selects a vaccine and marks it as "Received".
    3. System updates the digital record with clinic name and date.

2.3 Use Case: Log Chronic Vitals

- **Actor:** Chronic Patient.
- **Description:** Patient records daily vitals and receives alerts for dangerous levels.
- **Main Flow:**
    1. Patient enters current blood sugar or pressure readings.
    2. System saves the data and checks it against safety thresholds.
    3. System triggers a push alert if the reading is dangerous.

2.4. Use Case: Emergency Panic Alert

- **Actor:** Elderly User.
- **Description:** Elderly user sends their GPS location to the family during an emergency.
- **Main Flow:**

1. Elderly user presses the large Panic Button on the home screen.
2. System acquires real-time GPS coordinates.
3. System broadcasts a high-priority alert to all family members.

2.5 Use Case: Upload Medical Record

- **Actor:** Any Family Member.
- **Description:** User digitizes paper prescriptions or lab results for central storage.
- **Main Flow:**
  1. User captures a photo of the lab result or prescription.
  2. System uploads the compressed image to secure cloud storage.
  3. System saves the document into the user's digital health history

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3.Functional Requirements (FR):

FR-1: The system shall enable the admin to add multiple family members and assign them specific profile types (e.g., child, elderly, pregnant, chronic)
FR-2: The system shall support image uploads (JPEG/PNG) for lab results and prescriptions
FR-3: The system shall aggregate all family health events (tests, vaccines, appointments) into a single master calendar.
FR-4: The system shall generate a dynamic vaccination timeline based on the child's date of birth and Egyptian Ministry of Health (MOH) guidelines.
FR-5: The system shall allow parents to mark vaccines as "received" and input the clinic name and date of administration.
FR-6: The system shall provide a trimester-based checklist for mandatory prenatal tests, including blood tests and ultrasounds**.**
FR-7: The system shall allow pregnant users to log and confirm daily intake of prenatal supplements (e.g., folic acid, iron).
FR-8: The system shall offer a searchable, color-coded reference list (Green/Red) for safe vs. unsafe Egyptian local dishes
FR-9: The system shall send a push notification to the Family Admin if an elderly member misses a medication window.
FR-10: The system shall broadcast the user's real-time GPS coordinates to all registered family members when the panic button is pressed.
FR-11: The system shall allow chronic patients to log daily vital signs, specifically blood sugar and blood pressure and rigger an immediate push notification alert when a logged vital sign exceeds a predefined dangerous threshold
FR-12: The system shall allow chronic patients to log daily vital signs, specifically blood sugar and blood pressure.

FR-13: The system shall generate an automated monthly clinical summary (PDF) aggregating vitals and adherence data for doctors.

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4. Non-Functional Requirements (NFRs):

4.1 Security & Privacy (The Most Critical for Healthcare)

NFR-1: Data Encryption: All sensitive medical data, including chronic conditions and lab results, must be encrypted both at rest and during transit using industry-standard protocols (e.g., AES-256).

NFR-2: Authentication & Authorization: The system must enforce proper authentication using phone numbers and PINs to ensure only authorized family members can access private medical records.

NFR-3: Privacy Compliance and Secure Storage : The platform must ensure no exposure of private medical data to unauthorized third parties and must be stored in a secure cloud environment with restricted access.

4.2 Usability & User Experience (UX)

NFR-4: Minimalist Interface: The elderly user module must feature a deliberately minimal UI with large, readable buttons to accommodate users with limited technical skills.

NFR-5: Consistency: The application's design must match approved UI/UX mockups across all modules to ensure a cohesive user experience.

NFR-6: Accessibility: All text must be readable with proper labels and high-contrast ratios to support users with visual impairments.

NFR-7: User Feedback: The system must provide clear and immediate visual feedback for all user actions, including success messages, error alerts, and loading states.

4.3 Reliability & Availability

NFR-8: Emergency Availability: Critical safety features, such as the Panic Button and Vital-threshold alerts, must be available and operational at all times.

4.4 Performance & Scalability

NFR-9:Response Time: All features, including the shared family calendar and medical forms, must load and respond within an acceptable time frame to prevent user frustration.

NFR-10:Efficient Data Handling: The system must handle real-time synchronization of data across multiple family profiles without degrading performance.

4.5 Data Integrity & Validation

NFR-11:Zero Data Loss: The application must ensure that no data is lost during the submission or updating process, even during intermittent connectivity.

4.6 Maintainability & Deployment

NFR-12:integration Integrity: New features must be integrated into the main system without breaking existing functionalities.

5.Usability & UX Considerations

5.1 Visual Accessibility (Target: Elderly & Low-Vision Users)

To ensure the app is usable by elderly family members or those with visual impairments:

- **Typography:** We use a minimum font size of 16px for body text and 20px+ for critical alerts to ensure readability without strain.
- **Touch Targets:** All interactive elements (buttons, toggles) have a minimum hit area of **44x44 pixels** to accommodate users with reduced motor precision or hand tremors.
- **High Contrast:** We utilize a high-contrast color palette (e.g., dark text on a light background) to ensure clarity under different lighting conditions and for users with declining vision.

5.2 Error Prevention & Safety (Target: Medical Accuracy)

In a healthcare context, data entry errors are not just bugs; they are health risks.

- **Input Validation:** The system restricts data types (e.g., only numbers for blood pressure) and sets "sanity boundaries" (e.g., preventing a heart rate entry of 0 or 500) to maintain data integrity.

- **Confirmation Actions:** Critical actions, such as deleting a child's vaccination record or sending a Panic Alert, require a second confirmation tap to prevent accidental triggers.
- **Immediate Feedback:** When a user logs a vital sign, the system provides immediate color coded feedback (Green for "Normal," Red/Warning for "Requires Attention")

5.3 Visual "Sanity Checks" (Human-in-the-Loop)

Because health data is sensitive, we ensure the system does not act on potential typos without user verification

- . **Verification of Extremes**: Before the app broadcasts an emergency alert to the entire family (e.g., for a dangerously high blood sugar reading), it displays a verification screen: "You entered 500 mg/dL. This is a critical level. Is this correct?"
- **Impact**: This prevents the "Boy who cried wolf" effect, ensuring that family members only receive high-priority alerts for genuine health crises, not accidental typing errors.

5.4.Cognitive Load Reduction (Target: Ease of Use)

Healthcare management is stressful; the UI should remain calm and simple.

- **The 3-Click Rule:** Our information architecture ensures that any user can reach their specific module (Pregnancy, Child, or Elderly) within a maximum of three taps from the Home Screen.
- **Progressive Disclosure:** Instead of showing all medical history at once, we use "collapsible sections" so the user is not overwhelmed by information at a single glance.
- **Recognition over Recall:** We use standard medical iconography (e.g., a syringe for vaccines, a heart for vitals) so users can navigate intuitively without needing to learn new symbols.

5.5 Localized Context (Target: Egyptian Families):

- **Language:** The UI is designed to support Arabic and English, ensuring terminology is culturally and linguistically appropriate for all members of an Egyptian household.

- **Familiarity:** The "Digital Vaccination Booklet" design mimics the layout of the physical Ministry of Health booklet, making the transition from paper to digital easier for parents

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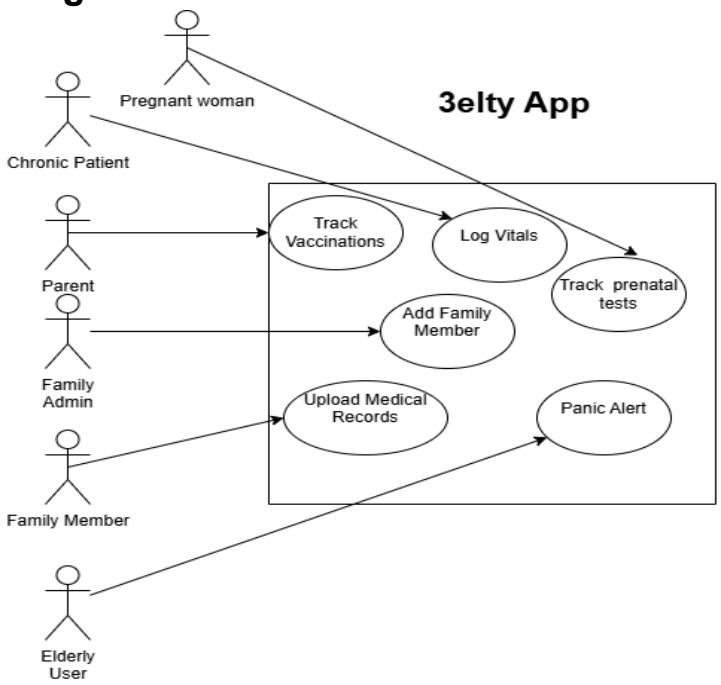
6. Diagrams

6.1 Use case diagram



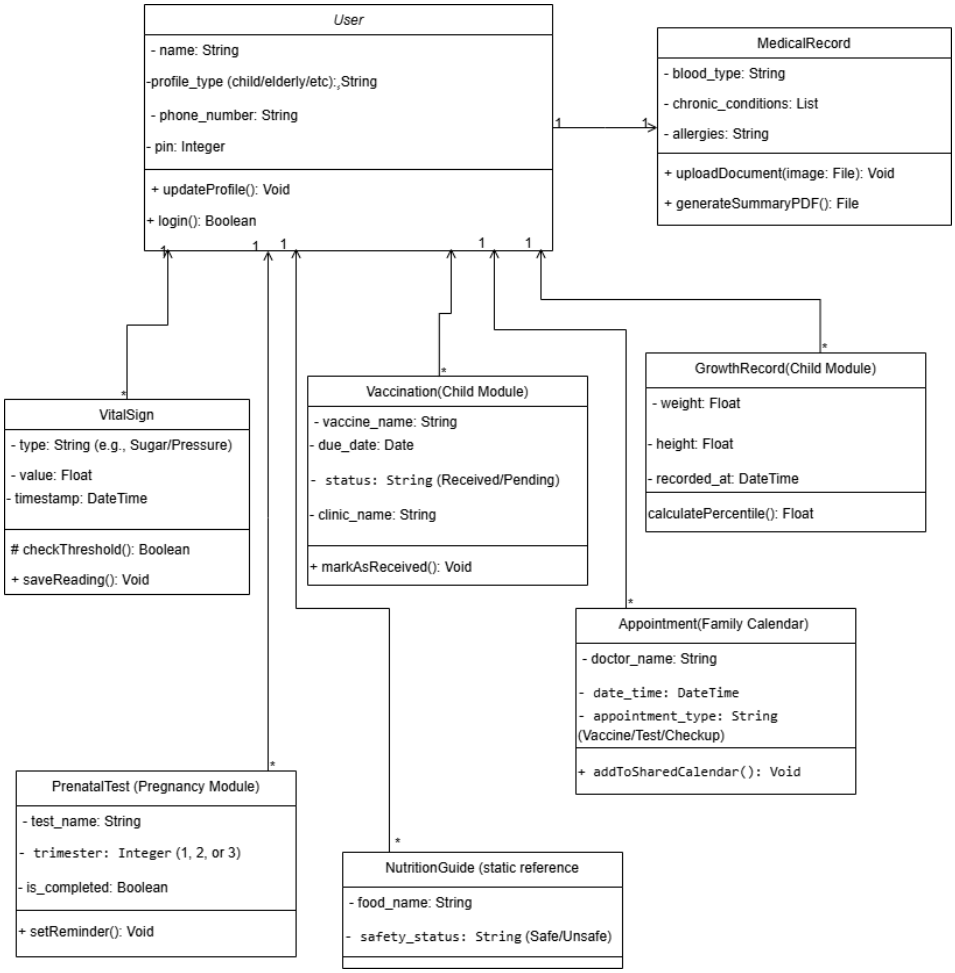
6.1.1 Actors (Users)

- **Pregnant woman:** Focuses on pregnancy-related tracking.
- **Chronic Patient:** Manages ongoing health conditions.
- **Parent:** Responsible for children's health records.
- **Family Admin:** Manages the family group and members.
- **Family Member:** General user of the system.
- **Elderly User:** Needs quick access to emergency features.

6.1.2 Key Use Cases (Features)

- **Track prenatal tests:** Monitoring pregnancy progress and scans.
- **Log Vitals:** Recording daily health metrics (e.g., blood pressure, sugar).
- **Track Vaccinations:** Keeping a schedule for immunization shots.
- **Add Family Member:** Administrative tool to build the family profile.
- **Upload Medical Records:** Digital storage for lab results and reports.
- **Panic Alert:** One-touch emergency notification for seniors.

6.2 Class diagram (core domain model)



6.2.1 Core Data

- **User:** Stores personal info and handles login.
- **MedicalRecord:** A 1-to-1 link to the User for static data (Blood type, allergie**s).**

6.2.2 Specialized Modules

The User can have multiple ($1..*$) entries in:

- **VitalSign:** For tracking sugar/pressure levels.
- **Vaccination & Growth:** (Child Module) To monitor shots and physical development.
- **PrenatalTest:** (Pregnancy Module) To track tests by trimester.
- **Appointment:** A shared family calendar for doctor visits.
- **NutritionGuide**: A reference for food safety.

6.2.3 Main Logic

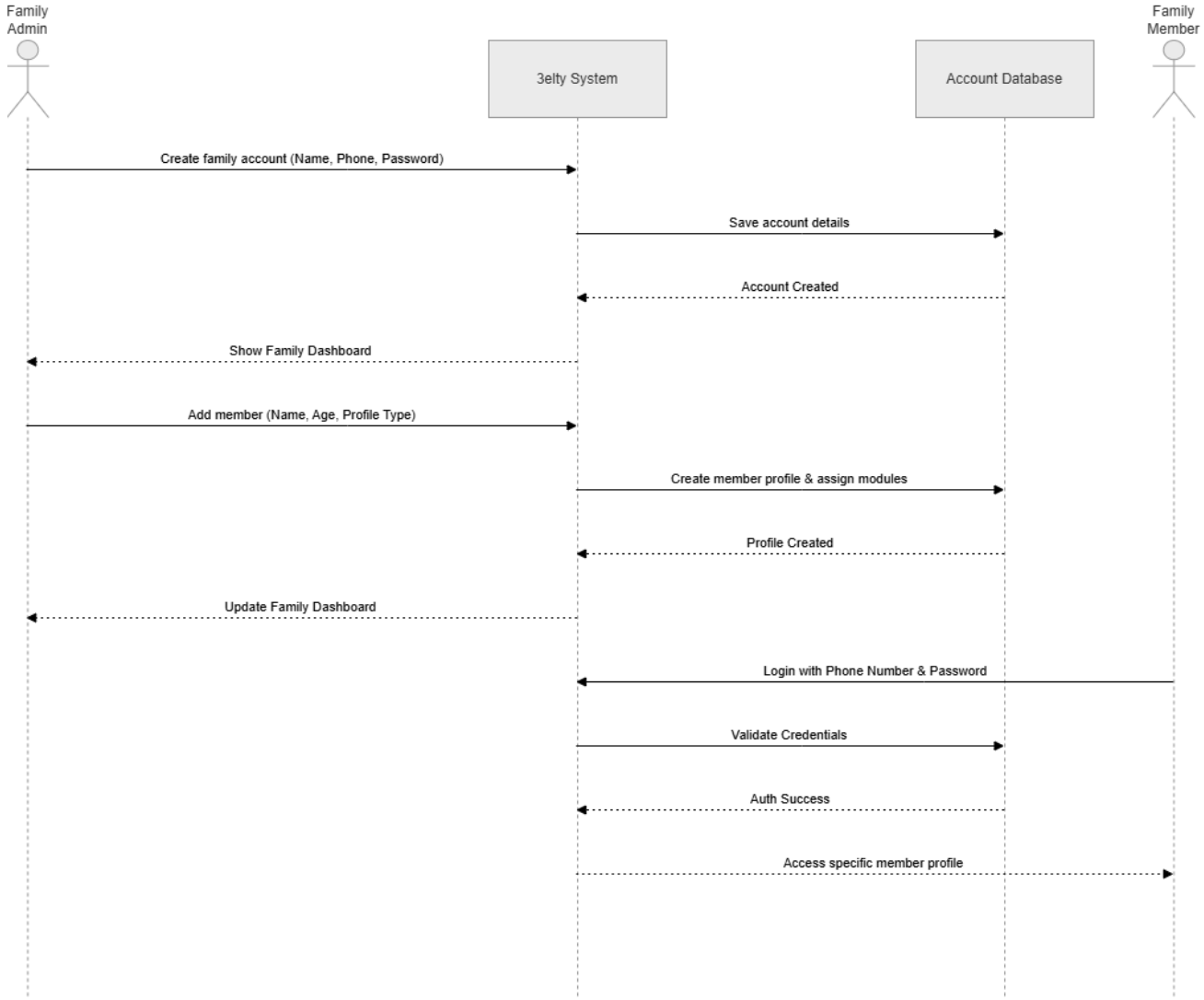
The system is User-centric, linking all medical logs, reminders, and alerts directly to an individual profile for easy tracking.

6.3 Sequence Diagrams: Core & Features

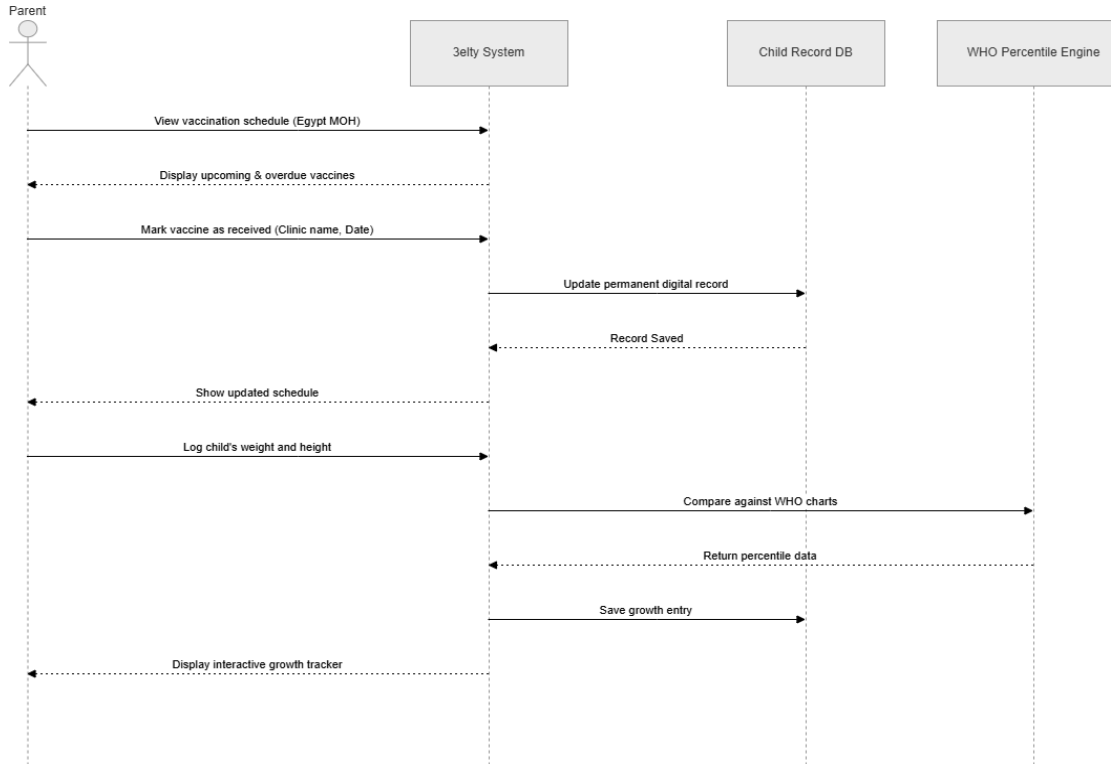
6.3.1The Core Purpose

The core function of these sequence diagrams is to bridge the gap between the user requirements (Jira Stories) and the backend technical implementation. They visualize the exact chronological flow of data, mapping out *what* happens, *when* it happens, and *which* system components are responsible for executing the action.

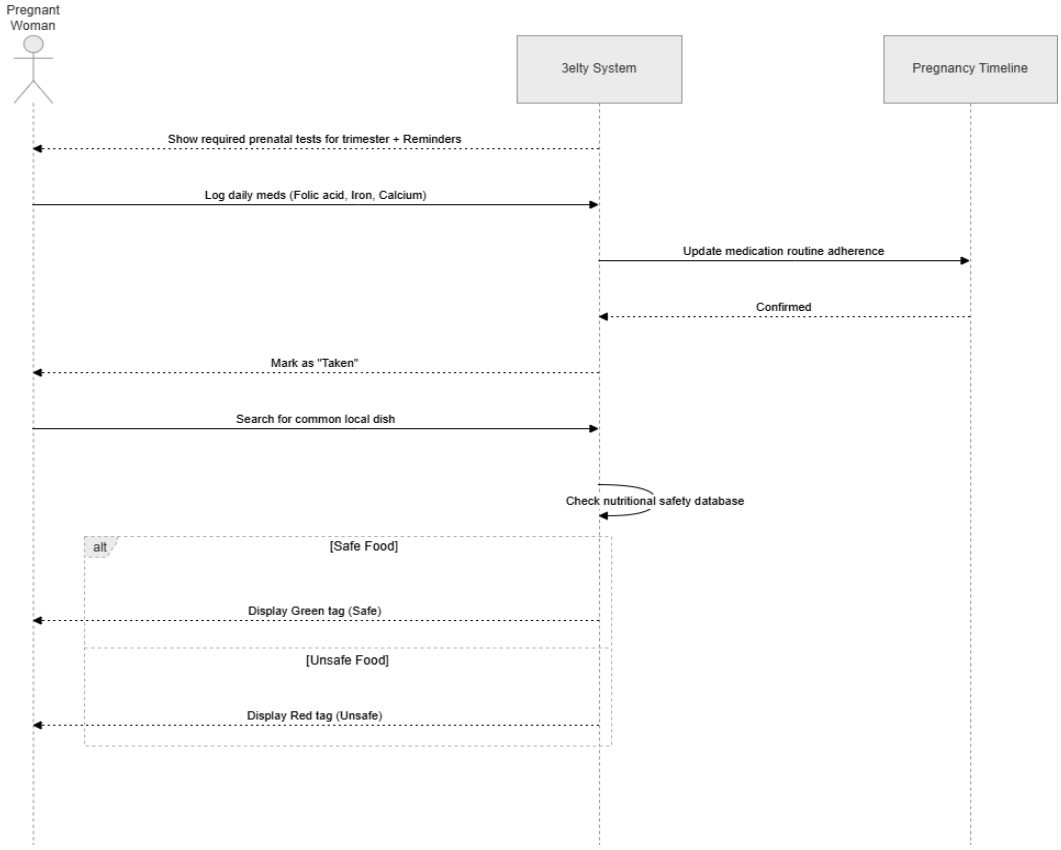
Authentication & Family Account



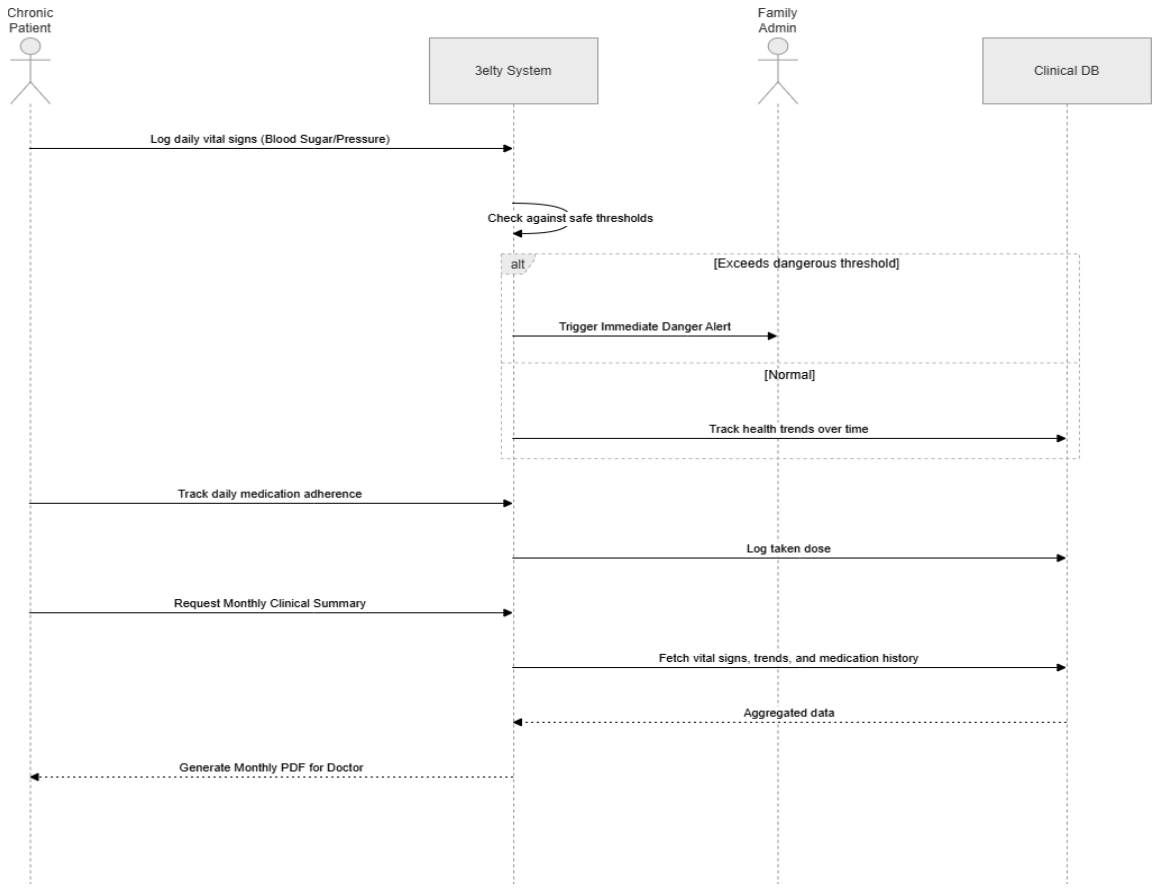
Child Module



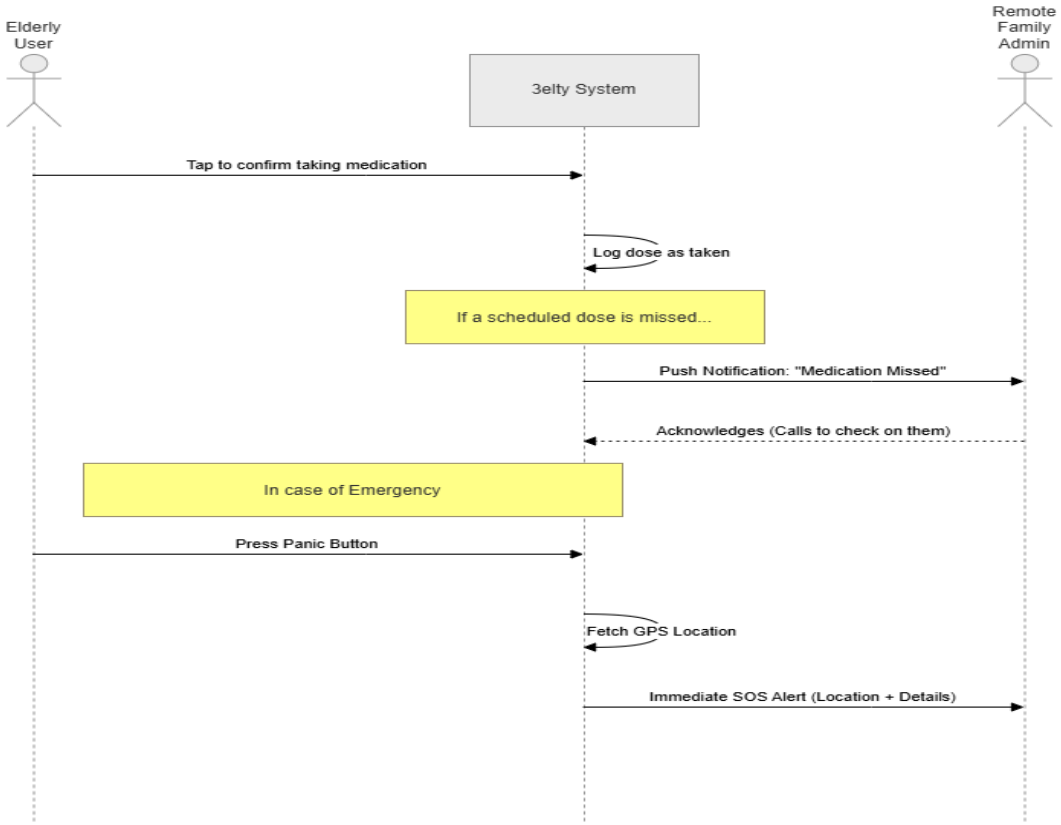
Pregnant Women Module



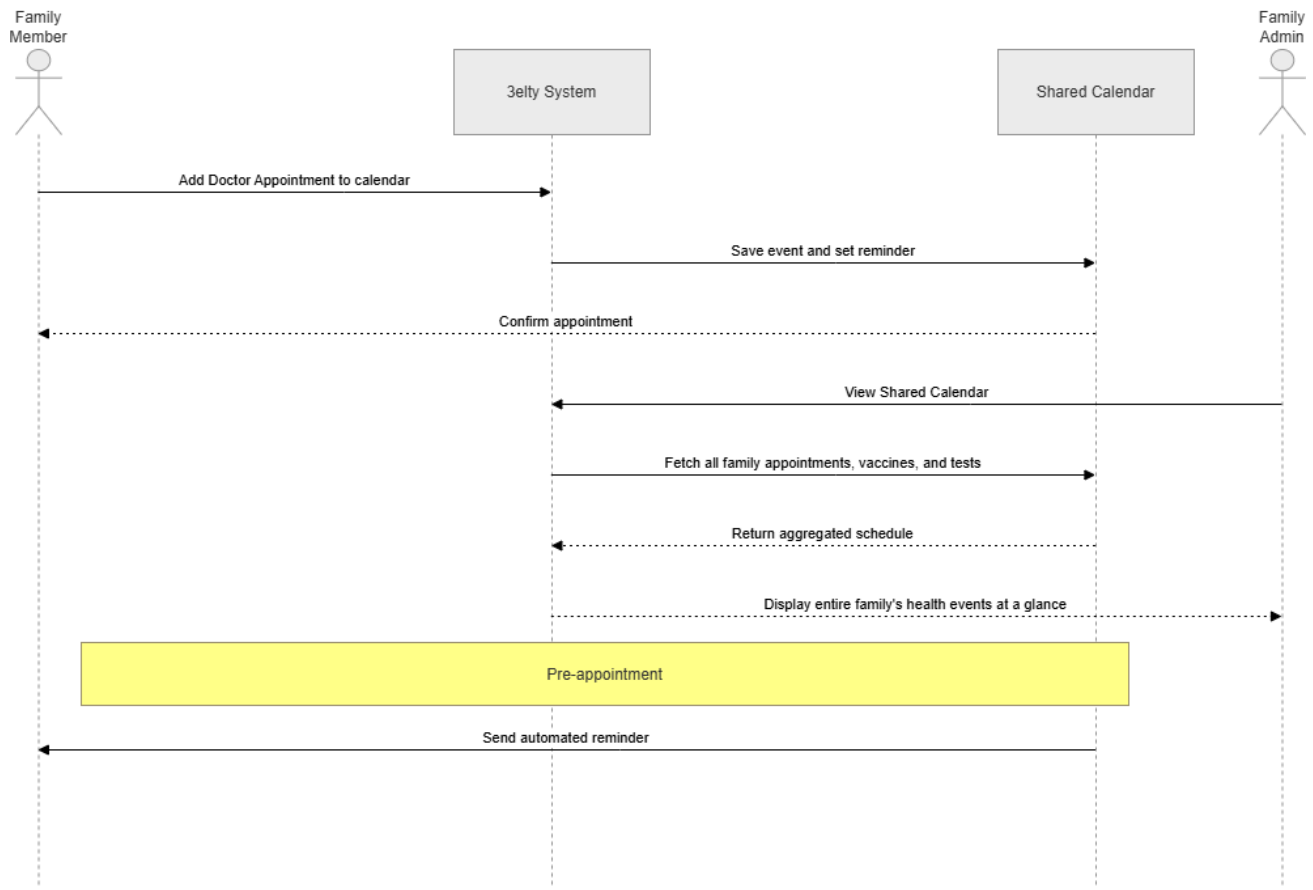
Chronic Patient Module



Elderly User Module



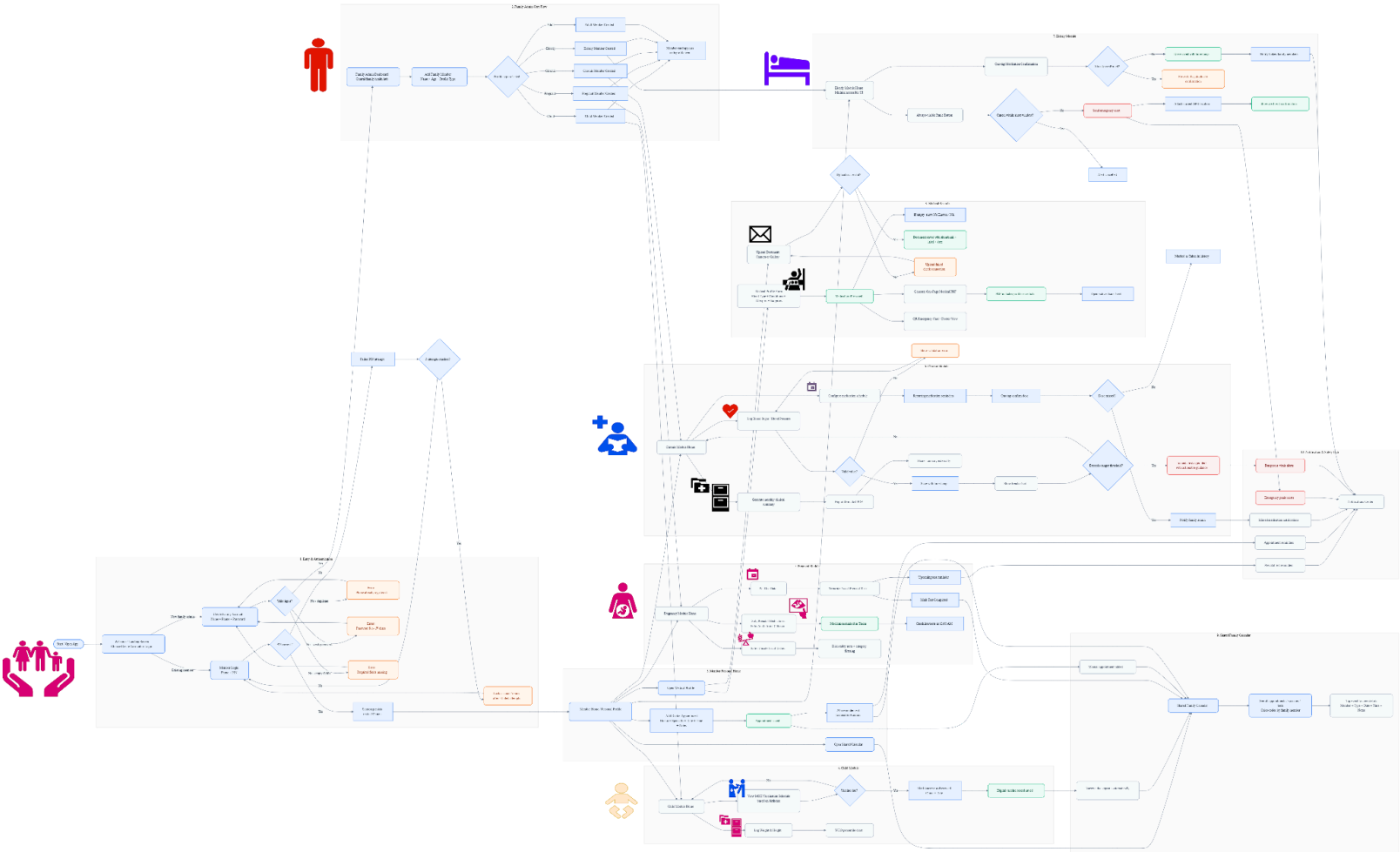
Family Calendar Module



6.3.2Key Diagram Features To ensure clear communication across the BitCare development team, these diagrams utilize the following key structural features:

- **Actors & Systems:** Distinct representation of human interactions (e.g., Family Admin, Elderly Patient) versus technical system boundaries (e.g., 3elty App, Clinical Database, WHO Percentile Engine).
- **Chronological Lifelines:** Time flows top-to-bottom, providing a clear visual timeline of synchronous and asynchronous events (like automated pre-appointment reminders).
- **Action Mapping:** Differentiates between active commands (solid arrows, like "Log Vital Sign") and system responses/callbacks (dashed arrows, like "Return Percentile Data")**.**
- **Conditional Logic (Alt Blocks):** Embedded "If/Then" routing to map out critical edge cases—such as triggering a high-priority Danger Alert *only if* a patient's vitals exceed safe thresholds, otherwise defaulting to a standard database log.

6.4 User flow diagram:



6.4.1 Core Features

- Tailored Health Profiles: Dedicated modules for different family members (Pregnancy, Pediatrics, Chronic Care, and Geriatrics) with customized interfaces.
- Centralized Family Coordination: A shared, color-coded calendar that aggregates all appointments, vaccinations, and medical tests in one view.
- Smart Medication Tracking: Daily checklist with an automated reset logic (at 12:00 AM) and proactive adherence reminders.
- Digital Medical Vault: Secure storage for medical records with the ability to generate "One-Page Medical PDFs" and QR Emergency Cards.
- Local Health Intelligence: Context-aware resources like the "Local Food Safety Guide" for pregnant users.

6.4.2 Safety & Emergency Hub

- Real-time Vital Alerts: Automated triggers that notify family members if dangerous health readings (blood pressure/sugar) are detected.
- One-Tap Emergency SOS: Instant GPS location sharing with the entire family network during critical medical crises.
- Proactive Reminders: Intelligent notification system for upcoming prenatal tests and missed critical medications.
- Data Privacy (PIN Lock): A secondary security layer (PIN Code) with a lockout mechanism after 3 failed attempts to protect sensitive health records.

6.4.3 Technical Flow & Input Handling

- Robust Input Validation: Strict logic to ensure data integrity (e.g., verifying email formats, preventing illogical date entries, or negative vital values).
- System State Persistence: Maintaining the "Taken/Not Taken" status of medications even if the app is closed, ensuring reliable progress tracking.
- Graceful Error Handling: Providing clear user feedback during technical failures (e.g., "Upload failed, check connection") to maintain system transparency.
- Conditional UI Rendering: Dynamic interface adaptation where the flow and tools change automatically based on the selected user profile type.